MIL-C-46627A(MU)
9 November 1964
SUPERSEDING
MIL-C-46627(MU)
26 January 1962

MILITARY SPECIFICATION
CATAPULT, AIRCRAFT EJECTION SEAT, M4A1, ASSEMBLY

1. SCOPE

1.1 This specification covers Catapult, Aircraft Ejection Seat, M4A1, Assembly for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MS-28741 - Hose Assembly-Detachable End Fitting, Medium Pressure

DRAWINGS

Munitions Command

F8593669 - Catapult, Aircraft Ejection Seat, M4A1, Assembly
F8797383 - Ammunition Container, Fiber M301 For Catapult, Aircraft Ejection Seat, M4A1 or M4
F8797384 - Box, Packing, Ammunition for Catapult, Aircraft Ejection Seat, M4A1 or M4 in Fiber Container M301
IEL-8593669 - Index to Inspection Equipment List for Catapult, Aircraft Ejection Seat, M4A1, Assembly

FSC 1375

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(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications.

3.2 Torque requirements.- Torque requirements shall be as specified in Table I. No loosening or deformation shall occur upon application of the breakaway torque.

Table I

Component	B Seating or Bottoming (Including A) (Inch Pounds)	C Breakaway (Inch Pounds)
Block and inside tube after insertion of cartridge	400 + 50	300 min.
Ring, stop and ring, tube stop	500 + 50	400 min.

3.3 Assembly

3.3.1 The flange face surface of the plug shall be in complete contact with the cartridge.

3.3.2 The block shall be free to rotate 360 degrees.

3.4 Residual magnetism.- The catapult shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.5 Functional

3.5.1 Ballistics.- Catapults loaded with accepted M37 cartridges shall propel a 325 pound mass vertically and shall comply with the requirements listed in Table II.

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Table II

Temperature	Velocity (min.)	Acceleration, Max. Pressure Time	Acceleration, Max. Load Cell	Rate of Change of Acceleration (max.)	Ignition Delay (max.) (see 6.2)
-65° to -70°F	33 fps	12g	10g	85g/sec	150 ms
65° to 75°F	38 fps	12.5g	10.5g	100g/sec	150 ms
160° to 165°F	41 fps	14g	14g	125g/sec	150 ms

3.5.2 The catapult shall not fail to fire.

3.5.3 The catapult shall function without mechanical failure (see 6.1).

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.- The component shall be free from cracks, splits, cold shuts, inclusions, porosity or similar defects.

3.6.2 Burrs.- The component shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.- The component or assembly shall not contain chips, dirt, grease, rust corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.2 Inspection provisions

4.2.1 Lot

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-414, as applicable.

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4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table III. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table III. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table III

Lot Size	No. of Samples	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3

4.2.2.1 Cartridge, Aircraft Ejection Seat, Catapult, M37 Dwg.D8593745, a detail of Dwg. F8593669.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defects	Visual
3. "O" ring missing	Visual
4. Sealing disc missing, damaged or improperly secured	Visual
5. Marking missing, incorrect or illegible	Visual
MAJOR	
101. "O" ring split, cut or cracked	Visual
102. Foreign matter	Visual
MINOR	
201. Case scratched, wrinkled, bulged or dented	Visual
202. Varnish missing or inadequate	Visual

4.2.2.2 Seal, Dwg C8593665, a detail of Dwg F8593669.

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Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Inside diameter under flange	Gage
202. Inside diameter adjacent to flange	Gage
203. Radius missing or incorrect	Visual
204. Metal defective	Visual
205. Protective finish inadequate or defective	Visual
206. Burr	Visual
207. Foreign matter	Visual

4.2.2.3 Assembly, Dwg. F8593669, less all dwgs. except C8593644, B8593654, C8593657, F8593672, C8593679 and AN 6227-11.

CRITICAL	
1. Gas port hole obstructed	Visual
2. Any part missing	Visual
3. Improper assembly	Visual
MAJOR	
101. Thread damaged	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

4.2.2.4 Assembly, Dwg. F8593669, less B8593660, D8593673, C8593676, C8593677, C8593678, D8593680, D8593684 and C8593741.

CRITICAL	
1. Flange face surface not in complete contact with cartridge	Gage
2. Cartridge missing	Visual
3. Seals inside tube missing	Visual

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Defect	Method of Inspection
MAJOR	
101. Shock washer missing	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

4.2.2.5 Assembly, Dwg. F8593669 less all drawings except D8593681 and C8593678.

CRITICAL	
None defined	
MAJOR	
101. Spacer not flush with telescoping tube	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

4.2.2.6 Assembly, Dwg. F8593669 less all drawings except B8593660, D8593674, C8593677, D8593684 and C8593741.

CRITICAL	
1. Any part missing	Visual
2. Improper assembly	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

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Defect	Method of Inspection
4.2.2.7 Assembly, Drawing F8593669	
CRITICAL	
1. Tubes not free to slide	Manual
2. Mixed catapults	Visual
3. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Total length, max.	Gage
102. Diameter of mounting hole	Gage
103. Length between mounting points	Gage
104. Block not free to rotate 360°	Manual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Location of labels, incorrect	Visual
202. Shipping plug missing	Visual
203. Seal missing or insecure	Visual
204. Protective finish inadequate or defective	Visual
4.2.2.8 Packing in Fiber Container, Dwg. F8797383.	
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

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Defect	Method of Inspection

4.2.2.9 Packing in Wooden Box, Dwg. F8797384

CRITICAL

None defined

MAJOR

None defined

MINOR

201.	Contents loose	Manual
202.	Hardware missing, loose, broken or improperly engaged	Visual/Manual
203.	Strapping missing, loose, broken or improperly assembled	Visual/Manual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking misleading or unidentifiable	Visual
207.	Number of units incorrect	Visual
208.	Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.- A breakaway torque shall be applied to each catapult threaded joint in 3.2. Failure of any catapult to comply with the requirements of 3.2 shall be cause for rejection of the catapult.

4.2.3.2 Residual magnetism.- Each catapult shall be tested for excessive residual magnetism. Failure of any catapult to comply with the requirements of 3.4 shall be cause for rejection of the catapult.

4.2.3.3 Functional.- The following tests shall be performed by the Government:

4.2.3.3.1 Ballistics.- Fourteen units shall be tested to determine compliance with the requirements of 3.5.1, 3.5.2, and 3.5.3. The units shall be temperature conditioned as provided in Table IV and 4.3.3.1.1. The variables sampling plan of MIL-STD-414, Example C-2, for AQL listed in Table IV shall be used for each sample of 10 units. Failure of any unit to comply with the requirements of 3.5.1, 3.5.2, and 3.5.3 or failure of a sample to meet the criteria of the variables sampling plan shall be cause for rejection of the lot.

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Table IV

Temperature (F)	No. of Samples	AQL
-65° to -70°	2	
65° to 75°	10	.65%
160° to 165°	2	

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8593669.

4.3 Test methods and procedures.

4.3.1 Breakaway torque.- A torque wrench set at the torque requirement specified in 3.2 shall be used to determine the adequacy of the joint. The assembly shall be secured in the fixture prior to application of the torque.

4.3.2 Residual magnetism.- The catapult shall be tested in an area free of local magnetic effects by placing the catapult 6 inches from and at the same height as the compass, in a north-south horizontal position (block end at south) with the compass set in an east-west heading as shown in Figure 1. The catapult shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 180 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass.

4.3.3 Functional.

4.3.3.1 Ballistics.- The catapult shall be rigidly supported in the fixture by the mounting points and fired with an M3A1 Initiator connected to the catapult with ten (10) feet of aircraft hose, size 4, as specified in MS-28741. The catapult shall comply with the requirements of 3.5. The catapult shall be tested within ten (10) minutes after removal from the conditioning chamber. When this time limit is exceeded, the catapult shall be reconditioned at the original temperature.

4.3.3.1.1 Temperature conditioning.- A standard M4A1 Catapult, loaded with a dummy M37 Cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Level A - Packing and marking shall be in accordance with drawings F8797383, F8797384.

5.2 Data cards.- Data card information shall be as specified in MIL-A-2550.

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6. NOTES

6.1 A "mechanical failure" is defined; as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 "Ignition delay" is defined as the time that elapses between the firing of the cartridge primer and the beginning of the continuous rise on the pressure time record or thrust time record.

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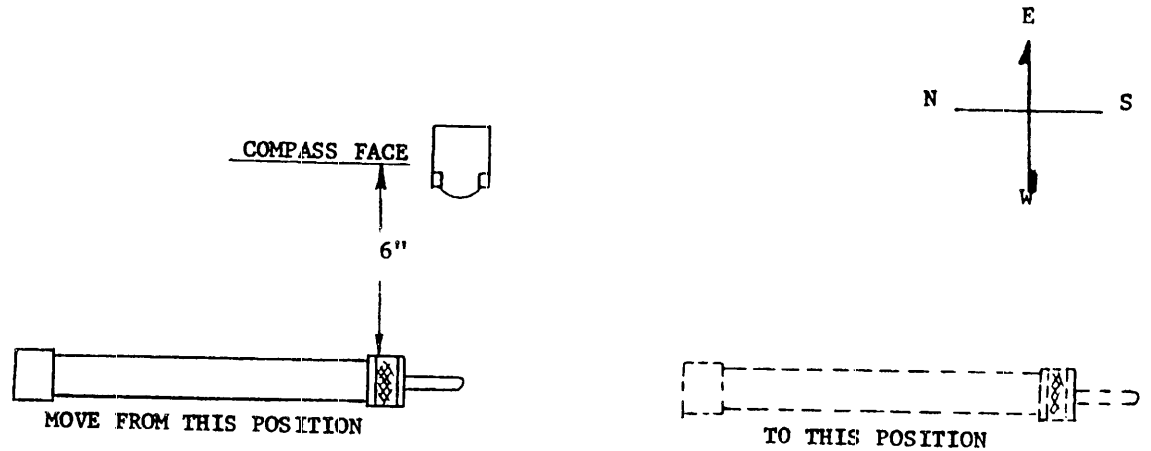
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PILOT'S STANDBY COMPASS, TYPE AN-5766



NOTE: PLAN VIEW

CATAPULT AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

FIGURE 1

LAYOUT FOR DETERMINATION OF RESIDUAL MAGNETISM IN CATAPULT, AIRCRAFT EJECTION SEAT, (M4A1)

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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